Artificial Intelligence Engineering Department

ALAYEN IRAQI UNIVERSITY AUIQ

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تحكم الي 4

ذكاء ثالثة

LECTURE - (4)
TRANSFER FUNCTION

Outline

- **Inverse Laplace Transform**
- **Partial fraction decomposition**
- **Impulse response**
- **Step response:**
  - **First-Order Prototype System**
  - **Second-Order Prototype System**

- The impulse response y(t) is therefore the *inverse Laplace transform* of the transfer function G(s), $y(t) = \mathcal{L}^{-1}[G(s)]$
- The general formula for computing the inverse Laplace transform is:

$$f(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st}ds$$

- Where σ is a real constant and is chosen larger than the real parts of all singular points of F(s).

In MATLAB use $\rightarrow \mathcal{L}^{-1}$ Inverse LaPlace transform

```
f = ILAPLACE(f)
```

Use helP for more Information. ⟶ MatlaP

MATLAB
» syms s
» F=2*s/(s^2+1)
» f=ilaplace(F)
f = 2*cos(t) ⟹ outPut

- The partial fraction decomposition of a rational function G(s) = N(s)/D(s) is (assuming $p_i \neq p_j$):

$$G(s) = \frac{\alpha_1}{s-p_1} + \cdots + \frac{\alpha_n}{s-p_n}$$

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- α_i is called the residue of G(s) in $p_i \in C$

$$\alpha_i = \lim_{s \to p_i}(s-p_i)G(s)$$

- The inverse Laplace transform of G(s) is easily computed by inverting each term:

$$\mathcal{L}^{-1}[G(s)] = \alpha_1 e^{p_1 t} + \cdots + \alpha_n e^{p_n t}$$

يكون تحويله سهل جدا !!

- For multiple poles p_i with multiplicity k we have the terms:

$$\frac{\alpha_{i1}}{(s-p_i)} + \cdots + \frac{\alpha_{ik}}{(s-p_i)^k}, \quad \alpha_{ij} = \frac{1}{(k-j)!}\lim_{s \to p_i}\frac{d^{(k-j)}}{ds^{(k-j)}}[(s-p_i)^k G(s)]$$

- and the inverse Laplace transform is :

$$\alpha_{i1}e^{p_i t} + \cdots + \alpha_{ik}\frac{t^{k-1}}{(k-1)!}e^{p_i t}$$

- Example:
- Consider the function : $G(s)=\dfrac{5s+3}{(s+1)(s+2)(s+3)}=\dfrac{5s+3}{s^3+6s^2+11s+6}$ الجزء الثالث
- which is written in the partial-fraction expanded form: $\dfrac{A}{s+1}+\dfrac{B}{s+2}+\dfrac{C}{s+3}$

$$G(s)=\frac{K_{-1}}{s+1}+\frac{K_{-2}}{s+2}+\frac{K_{-3}}{s+3}$$ كذلك الجبر

- The coefficients K_{-1}, K_{-2}, and K_{-3} are determined as follows:

$$K_{-1}=[(s+1)G(s)]\big|_{s=-1}=\frac{5(-1)+3}{(2-1)(3-1)}=-1$$

$$K_{-2}=[(s+2)G(s)]\big|_{s=-2}=\frac{5(-2)+3}{(1-2)(3-2)}=7$$

$$K_{-3}=[(s+3)G(s)]\big|_{s=-3}=\frac{5(-3)+3}{(1-3)(2-3)}=-6$$

$$G(s)=\frac{-1}{s+1}+\frac{7}{s+2}-\frac{6}{s+3}$$

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- Taking the inverse Laplace transform on both sides using the Laplace , we get:

$$g(t)=-e^{-t}+7e^{-2t}-6e^{-3t}\quad t\geq 0$$

ترجع لل t

- Example:
- Consider the function : $G(s)=\dfrac{1}{s(s+1)^3(s+2)}=\dfrac{1}{s^5+5s^4+9s^3+7s^2+2s}$
- G(s) is written as: $G(s)=\dfrac{K_0}{s}+\dfrac{K_{-2}}{s+2}+\dfrac{A_1}{s+1}+\dfrac{A_2}{(s+1)^2}+\dfrac{A_3}{(s+1)^3}$

$$K_0=[sG(s)]\big|_{s=0}=\frac{1}{2}$$

$$K_{-2}=[(s+2)G(s)]\big|_{s=-2}=\frac{1}{2}$$

$$A_3=[(s+1)^3G(s)]\big|_{s=-1}=-1$$

$$A_2=\frac{d}{ds}[(s+1)^3G(s)]\Big|_{s=-1}=\frac{d}{ds}\left[\frac{1}{s(s+2)}\right]\Big|_{s=-1}=0$$

$$A_1=\frac{1d^2}{2!ds^2}[(s+1)^3G(s)]\Big|_{s=-1}=\frac{1d^2}{2ds^2}\left[\frac{1}{s(s+2)}\right]\Big|_{s=-1}=-1$$

$$G(s)=\frac{1}{2s}+\frac{1}{2(s+2)}-\frac{1}{s+1}-\frac{1}{(s+1)^3}$$

$$g(t)=\mathcal{L}^{-1}[G(s)]$$

$$g(t)=\frac{1}{2}-e^{-t}+\frac{e^{-2t}}{2}-\frac{t^2e^{-t}}{2}$$

MATLAB

```
>> f5=1/(s*(s+1)^3*(s+2))
f5 =
1/(s*(s + 1)^3*(s + 2))
>> g=ilaplace(f5)
g =
exp(-2*t)/2 - exp(-t) -
   (t^2*exp(-t))/2 + 1/2
```

- **EXAMPLE:**
- **Simple Complex-Conjugate Poles**
- Consider the **second-order prototype function:** $G(s)=\dfrac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$
- **Let's consider** $\zeta<1$; ***so that the poles of*** $G(s)$ ***are complex***

$$G(s)=\frac{K_{-\sigma+j\omega}}{s+\sigma-j\omega}+\frac{K_{-\sigma-j\omega}}{s+\sigma+j\omega} \qquad \textbf{Where: } \sigma=\zeta\omega_n, \quad \omega=\omega_n\sqrt{1-\zeta^2}$$

$$\left.\begin{aligned} K_{-\sigma+j\omega}&=(s+\sigma-j\omega)G(s)\Big|_{s=-\sigma+j\omega}=\frac{\omega_n^2}{2j\omega}\\ K_{-\sigma-j\omega}&=(s+\sigma+j\omega)G(s)\Big|_{s=-\sigma-j\omega}=-\frac{\omega_n^2}{2j\omega}\end{aligned}\right\} G(s)=\frac{\omega_n^2}{2j\omega}\left[\frac{1}{s+\sigma-j\omega}-\frac{1}{s+\sigma+j\omega}\right]$$

- Taking the inverse Laplace transform on both sides of the last equation gives

$$g(t)=\frac{\omega_n^2}{2j\omega}e^{-\sigma t}\left(e^{j\omega t}-e^{-j\omega t}\right) \quad t\geq 0 \quad \textbf{Or:} \quad g(t)=\frac{\omega_n}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin\left(\omega_n\sqrt{1-\zeta^2}\,t\right)$$

- Remember that an input signal u(t) produces an output signal y(t) whose Laplace transform Y(s) is: $Y(s)=G(s)U(s)$
- where $U(s)=\mathcal{L}[u]$, for initial state $x(0)=0$
- **Speciale case:** impulsive input $u(t)=\delta(t)$, $U(s)=1$. The corresponding output y(t) is called the *impulse response.*
- G(s) is the Laplace transform of the impulse response y(t).

$$Y(s)=G(s)\cdot 1=G(s)$$

- **Example:**
  - *Integrator* $u(t)\rightarrow\boxed{\frac{1}{s}}\rightarrow y(t)$

$$u(t)=\delta(t)$$
$$y(t)=\mathcal{L}^{-1}[\tfrac{1}{s}]=\mathbb{I}(t)$$

  - *Double integrator* $u(t)\rightarrow\boxed{\frac{1}{s^2}}\rightarrow y(t)$

$$u(t)=\delta(t)$$
$$y(t)=\mathcal{L}^{-1}[\tfrac{1}{s^2}]=\mathbb{I}(t)t$$

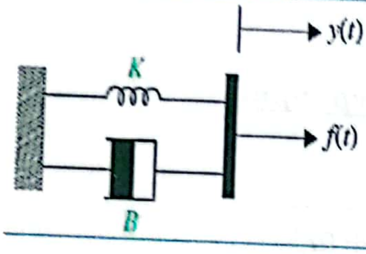
  - *Undamped oscillator* $u(t)\rightarrow\boxed{\frac{1}{s^2+1}}\rightarrow y(t)$

$$u(t)=\delta(t)$$
$$y(t)=\mathcal{L}^{-1}[\tfrac{1}{s^2+1}]=\mathbb{I}(t)\sin t$$

- ## First-Order Prototype System:
- The prototype forms of differential equations provide a common format of representing various components of a control system.

$$\frac{dy(t)}{dt}+\frac{1}{\tau}y(t)=\frac{1}{\tau}u(t)$$

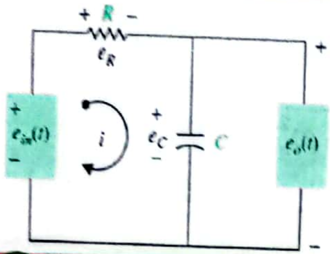
- where τ is known as the time constant of the system, which is a measure of how fast the system responds to initial conditions of external excitations.



$$\dot{y}(t)+\frac{K}{B}y(t)=\frac{Ku(t)}{B}$$

$\tau=\frac{B}{K}$

$Ku(t)=f(t)$ is the applied force to the system



$$\dot{e}_o(t)+\frac{1}{RC}e_o(t)=\frac{1}{RC}e_{in}(t)$$

$\tau=RC$ is the time constant

$e_{in}(t)$ is the input voltage

- ## First-Order Prototype System:
- For behaviors, we can solve Equation for a test input;

$$\frac{dy(t)}{dt}+\frac{1}{\tau}y(t)=\frac{1}{\tau}u(t)$$

- In this case, **a unit step input**:

$$u(t)=u_s(t)=\begin{cases}0, & t<0,\\ 1, & t\geq 0\end{cases}$$

- We can write the first order equation :

$$u_s(t)=\tau\frac{dy(t)}{dt}+y(t)$$

$$y(0)=\frac{dy(0)}{dt}=0,$$

$$L(y(t))=Y(s)$$

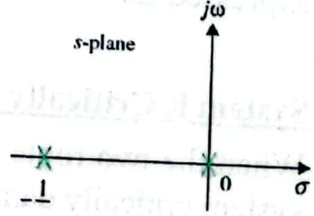
$$\mathcal{L}(u_s(t))=\frac{1}{s}$$

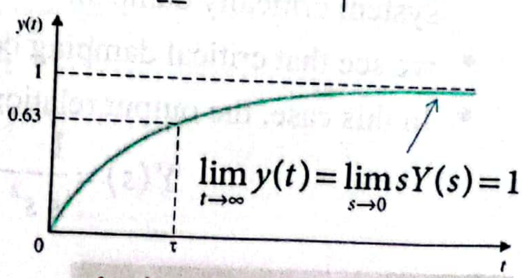
$$\frac{1}{s}=s\tau Y(s)+Y(s)$$

$$Y(s)=\frac{1}{s}\frac{1}{\tau s+1}$$

$$Y(s)=\frac{K_0}{s}+\frac{K_{-1/\tau}}{\tau s+1}$$

$$y(t)=1-e^{-t/\tau}$$





$$\lim_{t\to\infty}y(t)=\lim_{s\to 0}sY(s)=1$$

- **Second-Order Prototype System**

- The standard second-order prototype system has the form

$$\frac{d^2y(t)}{dt^2}+2\zeta\omega_n\frac{dy(t)}{dt}+\omega_n^2y(t)=\omega_n^2u(t)$$

- where ζ is known as the damping ratio, ω_n is the natural frequency of the system, y(t) is the output variable, and u(t) is the input.
- Damping ratio ζ plays an important role in the time response of the prototype second-order system.
- We can solve the Eq for a test **input-in this case a unit step input:**

$$u(t)=u_s(t)=\begin{cases}0, & t<0,\\ 1, & t\geq 0\end{cases}$$

$$y(0)=\frac{dy(0)}{dt}=0$$

$$\mathcal{L}(u_s(t))=U(s)=\frac{1}{s}$$

$$Y(s)=\frac{1}{s}\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

$$G(s)=\frac{Y(s)}{U(s)}=\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

- **Second-Order Prototype System**

$$G(s)=\frac{Y(s)}{U(s)}=\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

- The characteristic equation of the prototype second-order system is obtained by setting the denominator of Eq to zero:

$$\Delta(s)=s^2+2\zeta\omega_n s+\omega_n^2=0$$

- The two poles of the system are the roots of the characteristic equation, expressed as:

$$s_1,s_2=-\zeta\omega_n\pm\omega_n\sqrt{\zeta^2-1}$$

1- System is Critically Damped $\zeta=1$:

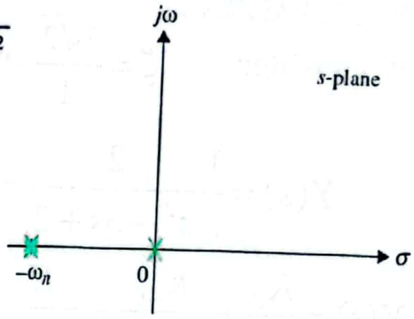
- When the two roots of the characteristic equation are real and equal, we call the system critically damped.
- we see that critical damping occurs when $\zeta=1$.
- In this case, the output relation in the s-domain, is rewritten as:

$$Y(s)=\frac{1}{s}\frac{\omega_n^2}{s^2+2\omega_n s+\omega_n^2}=\frac{1}{s}\frac{\omega_n^2}{(s+\omega_n)^2}$$

- **Second-Order Prototype System**

1- System Is Critically Damped $\zeta = 1$:

$$Y(s) = \frac{1}{s}\frac{\omega_n^2}{s^2 + 2\omega_n s + \omega_n^2} = \frac{1}{s}\frac{\omega_n^2}{(s+\omega_n)^2}$$



- The transfer function : $G(s) = \dfrac{\omega_n^2}{(s+\omega_n)^2}$
- where G(s) has two repeated poles at s = $-\omega_n$.
- Hence:

$$Y(s) = \frac{K_0}{s} + \frac{A_1}{(s+\omega_n)} + \frac{A_2}{(s+\omega_n)^2}$$

$$K_0 = \left[(s)\frac{1}{s}\frac{\omega_n^2}{(s+\omega_n)^2}\right]\Bigg|_{s=0} = 1$$

$$A_1 = \frac{d}{ds}\left[(s+\omega_n)^2\frac{1}{s}\frac{\omega_n^2}{(s+\omega_n)^2}\right]\Bigg|_{s=-\omega_n} = -1$$

$$A_2 = \left[(s+\omega_n)^2\frac{1}{s}\frac{\omega_n^2}{(s+\omega_n)^2}\right]\Bigg|_{s=-\omega_n} = -1$$

$$Y(s) = \frac{1}{s} - \frac{1}{(s+\omega_n)} - \frac{1}{(s+\omega_n)^2}$$

Taking the inverse Laplace transform:

$$y(t) = 1 - e^{-\omega_n t} - te^{-\omega_n t} \quad t \geq 0$$

- **Second-Order Prototype System**

2- System Is Over Damped $\zeta > 1$:

- When the two roots of the characteristic equation are distinct and real, we call the system overdamped.
- we see that an overdamped scenario occurs when $\zeta > 1$.
- In this case:

$$Y(s) = \frac{1}{s}\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- G(s) has two poles:

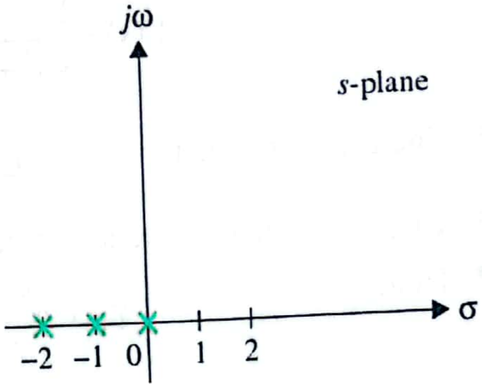
$$s_1, s_2 = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

suppose $\sigma = \zeta\omega_n$ and $\omega = \omega_n\sqrt{\zeta^2 - 1}$

• Second-Order Prototype System

2- System Is Over Damped $\zeta > 1$:

- **EXAMPLE:**
- **consider** $\zeta = \frac{3\sqrt{2}}{4}$ and $\omega_n = \sqrt{2}$ rad/s:



$$Y(s) = \frac{1}{s}\frac{2}{s^2+3s+2} = \frac{1}{s}\frac{2}{(s+1)(s+2)}$$

$$Y(s) = \frac{K_0}{s} + \frac{K_{-1}}{(s+1)} + \frac{K_{-2}}{(s+2)}$$

$$K_0 = \left[(s)\frac{1}{s}\frac{2}{(s+1)(s+2)}\right]\Bigg|_{s=0} = 1$$

$$K_{-1} = \left[(s+1)\frac{1}{s}\frac{2}{(s+1)(s+2)}\right]\Bigg|_{s=-1} = -2$$

$$K_{-2} = \left[(s+2)\frac{1}{s}\frac{2}{(s+1)(s+2)}\right]\Bigg|_{s=-2} = 1$$

$$Y(s) = \frac{1}{s} - \frac{2}{(s+1)} + \frac{1}{(s+2)}$$

$$y(t) = 1 - 2e^{-t} + e^{-2t} \qquad t \geq 0$$

• Second-Order Prototype System

3- System Is Under Damped $\zeta < 1$:

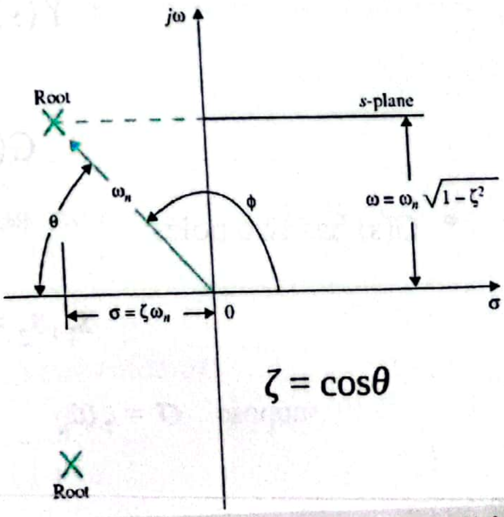
- When the two roots of the characteristic equation are complex with equal negative real parts, we call the system underdamped.

$$Y(s) = \frac{1}{s}\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2} \Rightarrow G(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

$$s_1, s_2 = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

Where : $\sigma = \zeta\omega_n$ and $\omega = \omega_n\sqrt{1-\zeta^2}$

- ω_n is the radial distance from the roots to the origin of the s-plane, $\omega_n = \sqrt{(\zeta\omega_n)^2 + \omega_n^2(1-\zeta^2)}$
- σ is the real part of the roots.
- ω is the imaginary part of the roots.
- ζ is the cosine of the angle between the radial line to the roots and the negative axis when the roots are in the left-half s-plane, or $\zeta = \cos\theta$



Second-Order Prototype System

3- System Is Under Damped $\zeta < 1$:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$Y(s) = \frac{K_0}{s} + \frac{K_{-\sigma+j\omega}}{s+\sigma-j\omega} + \frac{K_{-\sigma-j\omega}}{s+\sigma+j\omega}$$

$$K_0 = sY(s)\big|_{s=0} = 1$$

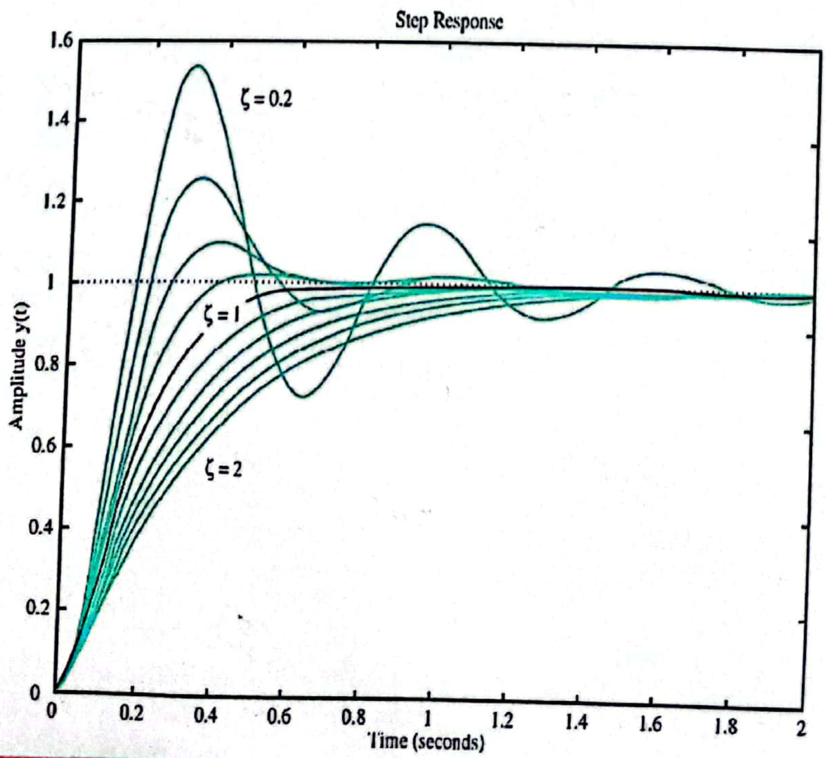
$$K_{-\sigma+j\omega} = (s+\sigma-j\omega)Y(s)\big|_{s=-\sigma+j\omega} = \frac{e^{-j\phi}}{2j\sqrt{1-\zeta^2}}$$

$$K_{-\sigma-j\omega} = (s+\sigma+j\omega)Y(s)\big|_{s=-\sigma-j\omega} = \frac{-e^{-j\phi}}{2j\sqrt{1-\zeta^2}}$$

$$\phi = \pi - \cos^{-1}\zeta$$

$$y(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin\left[\left(\omega_n\sqrt{1-\zeta^2}\right)t + \cos^{-1}\zeta\right] \quad t \geq 0$$

Second-Order Prototype System- Final Observations



End of Lecture 4

From Lec3 refrence to matlap /help

ζ zeta :- $G(s) = \dfrac{\omega^2}{s^2 + 2\zeta\omega s + \omega^2}$

المعادلة المميزة للمقام

اذا كان معامل التخامد كبير يعني الحركة اكثر استقرارا

اذا كان معامل التخامد صغير يعني الحركة ليس مستقرا "Struggling"

$S_1 = \dfrac{-B + \sqrt{\Delta}}{2A}$

$\Delta = B^2 - 4AC$

فقط بالمعادلة العامة "عموما"

$= - \dots \sqrt{\zeta^2 - 1}$

معامل التخامد

اذا كانت $\zeta < 1$ يعني $\Delta < 0$ يعني قسم تخيلي وقسم حقيقي